

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the canonical TE Curve Verification Pipeline offline comparison between RCIM Model-Bank Reproduction , recovered original, retuned paper-reference model banks, and repository-owned Wave 1 and Wave 2.1 model candidates. It starts from the current direction-aware comparison matrix.

Dataset And Split

- dataset config: config/datasets/transmission_error_dataset.yaml ;
- dataset root: data\simplified_dataset ;
- comparison mode: full_directional_candidate_matrix ;
- candidate count: 166 ;
- held-out curve count before candidate filtering: 194 ;
- percentage-error denominator: peak_to_peak_truth ;
- Fw candidates are evaluated only on forward curves;
- Bw candidates are evaluated only on backward curves;
- global candidates are evaluated on both directions and reported with direction-separated metrics.

Candidate Inventory

Candidate	Family	Source	Kind	Surface	Valid Directions
SVM19_Fw	SVM	rcim_track1	track1_reference_bank	Fw	forward
SVM19_Bw	SVM	rcim_track1	track1_reference_bank	Bw	backward
MLP19_Fw	MLP	rcim_track1	track1_reference_bank	Fw	forward
MLP19_Bw	MLP	rcim_track1	track1_reference_bank	Bw	backward
RF19_Fw	RF	rcim_track1	track1_reference_bank	Fw	forward
RF19_Bw	RF	rcim_track1	track1_reference_bank	Bw	backward
DT19_Fw	DT	rcim_track1	track1_reference_bank	Fw	forward
DT19_Bw	DT	rcim_track1	track1_reference_bank	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
ET19_Fw	ET	rcim_track1	track1_reference_bank	Fw	forward
ET19_Bw	ET	rcim_track1	track1_reference_bank	Bw	backward
ERT19_Fw	ERT	rcim_track1	track1_reference_bank	Fw	forward
ERT19_Bw	ERT	rcim_track1	track1_reference_bank	Bw	backward
GBM19_Fw	GBM	rcim_track1	track1_reference_bank	Fw	forward
GBM19_Bw	GBM	rcim_track1	track1_reference_bank	Bw	backward
HGBM19_Fw	HGBM	rcim_track1	track1_reference_bank	Fw	forward
HGBM19_Bw	HGBM	rcim_track1	track1_reference_bank	Bw	backward
XGBM19_Fw	XGBM	rcim_track1	track1_reference_bank	Fw	forward
XGBM19_Bw	XGBM	rcim_track1	track1_reference_bank	Bw	backward
LGBM19_Fw	LGBM	rcim_track1	track1_reference_bank	Fw	forward
LGBM19_Bw	LGBM	rcim_track1	track1_reference_bank	Bw	backward
ELM19_Fw	ELM	rcim_track1	track1_reference_bank	Fw	forward
ELM19_Bw	ELM	rcim_track1	track1_reference_bank	Bw	backward
rcim_original_SVM19_Fw	SVM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_MLP19_Fw	MLP	rcim_original	track1_reference_bank	Fw	forward
rcim_original_RF19_Fw	RF	rcim_original	track1_reference_bank	Fw	forward
rcim_original_DT19_Fw	DT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ET19_Fw	ET	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ERT19_Fw	ERT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_GBM19_Fw	GBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_HGBM19_Fw	HGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_XGBM19_Fw	XGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_LGBM19_Fw	LGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ELM19_Fw	ELM	rcim_original	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Fw	SVM	rcim_retuned	track1_reference_bank	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
rcim_retuned_MLP19_Fw	MLP	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_RF19_Fw	RF	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_DT19_Fw	DT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ET19_Fw	ET	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ERT19_Fw	ERT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_GBM19_Fw	GBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_HGBM19_Fw	HGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_XGBM19_Fw	XGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_LGBM19_Fw	LGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ELM19_Fw	ELM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Bw	SVM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_MLP19_Bw	MLP	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_RF19_Bw	RF	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_DT19_Bw	DT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ET19_Bw	ET	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ERT19_Bw	ERT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_GBM19_Bw	GBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_HGBM19_Bw	HGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_XGBM19_Bw	XGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_LGBM19_Bw	LGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ELM19_Bw	ELM	rcim_retuned	track1_reference_bank	Bw	backward
temporal_convolution_global	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
temporal_convolution_Fw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
temporal_convolution_Bw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
gru_sequence_global	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
gru_sequence_Fw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
gru_sequence_Bw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
lstm_sequence_global	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
lstm_sequence_Fw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
lstm_sequence_Bw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
feedforward_global	feedforward	wave1	wave1_exported_model	global	forward, backward
feedforward_Fw	feedforward	wave1	wave1_exported_model	Fw	forward
feedforward_Bw	feedforward	wave1	wave1_exported_model	Bw	backward
harmonic_regression_global	harmonic_regression	wave1	wave1_exported_model	global	forward, backward
harmonic_regression_Fw	harmonic_regression	wave1	wave1_exported_model	Fw	forward
harmonic_regression_Bw	harmonic_regression	wave1	wave1_exported_model	Bw	backward
periodic_mlp_global	periodic_mlp	wave1	wave1_exported_model	global	forward, backward
periodic_mlp_Fw	periodic_mlp	wave1	wave1_exported_model	Fw	forward
periodic_mlp_Bw	periodic_mlp	wave1	wave1_exported_model	Bw	backward
residual_harmonic_mlp_global	residual_harmonic_mlp	wave1	wave1_exported_model	global	forward, backward
residual_harmonic_mlp_Fw	residual_harmonic_mlp	wave1	wave1_exported_model	Fw	forward
residual_harmonic_mlp_Bw	residual_harmonic_mlp	wave1	wave1_exported_model	Bw	backward
tree_global	tree	wave1	wave1_exported_model	global	forward, backward
tree_Fw	tree	wave1	wave1_exported_model	Fw	forward
tree_Bw	tree	wave1	wave1_exported_model	Bw	backward
paper_original_best_Fw	best_composite	rcim_original	composite_reference_bank	Fw	forward
paper_retuned_best_Fw	best_composite	rcim_retuned	composite_reference_bank	Fw	forward
track1_best_Fw	best_composite	rcim_track1	composite_reference_bank	Fw	forward
paper_retuned_best_Bw	best_composite	rcim_retuned	composite_reference_bank	Bw	backward
track1_best_Bw	best_composite	rcim_track1	composite_reference_bank	Bw	backward
periodic_temporal_convolution_global	periodic_temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
periodic_temporal_convolution_Fw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_temporal_convolution_Bw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_gru_sequence_global	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_gru_sequence_Fw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_gru_sequence_Bw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_lstm_sequence_global	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_lstm_sequence_Fw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_lstm_sequence_Bw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_sparse_rcim_Fw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_sparse_rcim_Bw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense240_Fw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense240_Bw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense360_Fw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense360_Bw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
residual_harmonic_lstm_sequence_sparse_rcim_Fw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_sparse_rcim_Bw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense240_Fw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense240_Bw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense360_Fw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense360_Bw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
sequential_residual_offset_probe_global	sequential_residual_offset_probe	track2f_offset_aware_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_clean_sequential_residual_offset_global	track2f_bis_clean_sequential_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_harmonic_residual_offset_global	track2f_bis_harmonic_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_pointwise_control_global	track2g_curve_aware_pointwise_control	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_centered_shape_global	track2g_curve_aware_raw_centered_shape	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_offset_global	track2g_curve_aware_raw_offset	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_full_curve_composite_global	track2g_curve_aware_full_curve_composite	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2h_mae_robust_global	track2h_mae_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_smooth_l1_robust_global	track2h_smooth_l1_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
track2h_log_cosh_robust_global	track2h_log_cosh_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_quantile_p10_p50_p90_global	track2h_quantile_p10_p50_p90	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_gaussian_nll_global	track2h_gaussian_nll	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k2_global	track2h_mdn_k2	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k3_global	track2h_mdn_k3	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_l_gru_offset_residual_global	track2h_l_gru_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
track2h_l_causal_tcn_offset_residual_global	track2h_l_causal_tcn_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
wave3_harmonic_prior_residual_pointwise_control_global	wave3_harmonic_prior_residual_pointwise_control	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward
wave3_harmonic_prior_residual_smooth_l1_structured_global	wave3_harmonic_prior_residual_smooth_l1_structured	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward
wave52b_offset_centered_shape_harmonic_global	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	global	forward, backward
polished_feedforward_global	feedforward	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_harmonic_regression_global	harmonic_regression	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_global	periodic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_mlp_global	residual_harmonic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_tree_global	tree	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_harmonic_global	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_temporal_convolution_global	temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_gru_sequence_global	gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_lstm_sequence_global	lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_temporal_convolution_global	periodic_temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_gru_sequence_global	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_lstm_sequence_global	periodic_lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_1_sequential_residual_offset_probe_global	wave3_1_sequential_residual_offset_probe	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_clean_sequential_residual_offset_global	wave3_2_clean_sequential_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_harmonic_residual_offset_global	wave3_2_harmonic_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_curve_aware_pointwise_control_global	wave3_3_curve_aware_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_raw_centered_shape_curve_aware_global	wave3_3_raw_centered_shape_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_raw_offset_curve_aware_global	wave3_3_raw_offset_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_wave3_3_full_curve_composite_global	wave3_3_full_curve_composite	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_mae_robust_loss_global	wave4_1_mae_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_smooth_l1_robust_loss_global	wave4_1_smooth_l1_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_log_cosh_robust_loss_global	wave4_1_log_cosh_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_quantile_p10_p50_p90_global	wave4_2_quantile_p10_p50_p90	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_gaussian_nll_global	wave4_2_gaussian_nll	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k2_global	wave4_3_mixture_density_k2	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k3_global	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_gru_latent_offset_residual_global	wave4_4_gru_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_causal_tcn_latent_offset_residual_global	wave4_4_causal_tcn_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_pointwise_control_global	wave5_1_harmonic_prior_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	wave5_1_harmonic_prior_smooth_l1_structured	polished_model_development_registry	wave1_registry_model	global	forward, backward

Best Composite Reference Models

These candidates combine the approved best harmonic-wise cells into one TE Curve Verification Pipeline curve-reconstruction candidate. They are also repeated inside the source-group tables below, but this section keeps the composed models explicit.

Candidate	Source	Surface	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
paper_original_best_Fw	rcim_original	Fw	forward	0.002769	0.002951	6.250	13.827
paper_retuned_best_Fw	rcim_retuned	Fw	forward	0.001839	0.002041	4.109	9.866
track1_best_Fw	rcim_track1	Fw	forward	0.003014	0.003204	6.819	11.638
paper_retuned_best_Bw	rcim_retuned	Bw	backward	0.003675	0.004284	7.572	15.645

Candidate	Source	Surface	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track1_best_Bw	rcim_track1	Bw	backward	0.005027	0.005212	11.860	48.106

Forward Comparison

Original Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_original_ERT19_Fw	0.001471	0.001677	3.253	6.145
rcim_original_RF19_Fw	0.001767	0.001971	3.940	6.872
rcim_original_LGBM19_Fw	0.001801	0.002004	4.017	10.054
rcim_original_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_original_GBM19_Fw	0.001921	0.002122	4.312	8.193
rcim_original_HGBM19_Fw	0.002011	0.002217	4.493	10.617
rcim_original_ET19_Fw	0.002232	0.002432	4.985	11.357
rcim_original_XGBM19_Fw	0.002594	0.002814	5.805	10.574
paper_original_best_Fw	0.002769	0.002951	6.250	13.827
rcim_original_SVM19_Fw	0.003052	0.003324	6.767	13.827
rcim_original_ELM19_Fw	0.005423	0.005731	12.130	28.721
rcim_original_MLP19_Fw	0.018754	0.022589	42.943	107.515

Retuned Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912
rcim_retuned_RF19_Fw	0.001487	0.001699	3.292	5.998
rcim_retuned_ERT19_Fw	0.001807	0.002010	4.039	7.599
paper_retuned_best_Fw	0.001839	0.002041	4.109	9.866
rcim_retuned_HGBM19_Fw	0.001851	0.002056	4.126	9.401
rcim_retuned_LGBM19_Fw	0.001851	0.002055	4.135	9.866
rcim_retuned_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_retuned_ET19_Fw	0.002001	0.002196	4.426	9.533
rcim_retuned_XGBM19_Fw	0.002054	0.002264	4.588	10.488
rcim_retuned_SVM19_Fw	0.003167	0.003428	7.035	12.971
rcim_retuned_ELM19_Fw	0.007182	0.007463	16.181	40.024
rcim_retuned_MLP19_Fw	0.016647	0.020154	38.510	86.197

RCIM Model-Bank Reproduction Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
RF19_Fw	0.002164	0.002371	4.841	11.061
ERT19_Fw	0.002295	0.002491	5.163	13.141
GBM19_Fw	0.002779	0.002981	6.238	12.995
track1_best_Fw	0.003014	0.003204	6.819	11.638
DT19_Fw	0.003122	0.003313	7.011	13.928
SVM19_Fw	0.003236	0.003487	7.185	11.841
HGBM19_Fw	0.003251	0.003464	7.315	13.802
ET19_Fw	0.003267	0.003467	7.339	14.670
XGBM19_Fw	0.004190	0.004396	9.407	22.660
LGBM19_Fw	0.006812	0.007009	15.415	30.398
ELM19_Fw	0.007281	0.007573	16.352	36.326
MLP19_Fw	0.038690	0.047157	89.567	201.437

Wave 1 Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_global	0.002998	0.003364	6.590	11.601
tree_Fw	0.003053	0.003395	6.731	11.995
periodic_mlp_Fw	0.003366	0.003794	7.467	12.904
periodic_mlp_global	0.003380	0.003791	7.493	14.349
harmonic_regression_Fw	0.003394	0.003807	7.534	11.391
residual_harmonic_mlp_Fw	0.003434	0.003876	7.618	13.617
feedforward_Fw	0.003466	0.003904	7.685	13.132
feedforward_global	0.003550	0.003979	7.916	16.195
residual_harmonic_mlp_global	0.003649	0.004051	8.123	14.713
harmonic_regression_global	0.018314	0.018547	41.749	79.404

Wave 2.1 Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_lstm_sequence_global	0.002726	0.002959	6.142	14.092
periodic_gru_sequence_global	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_Fw	0.003186	0.003438	7.077	11.974
periodic_lstm_sequence_Fw	0.003266	0.003550	7.258	11.961
gru_sequence_Fw	0.003330	0.003762	7.378	13.029
periodic_temporal_convolution_Fw	0.003335	0.003708	7.404	12.518
lstm_sequence_Fw	0.003366	0.003800	7.450	11.807
periodic_temporal_convolution_global	0.003407	0.003724	7.581	14.480
lstm_sequence_global	0.003445	0.003863	7.642	12.032
temporal_convolution_global	0.003508	0.003928	7.792	14.045
gru_sequence_global	0.003546	0.003975	7.869	14.344
temporal_convolution_Fw	0.003603	0.004031	8.028	14.674

Wave 2.3 Residual Harmonic Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_gru_sequence_sparse_rcim_Fw	0.003194	0.003499	7.083	13.041
residual_harmonic_lstm_sequence_sparse_rcim_Fw	0.003229	0.003533	7.164	12.682
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003245	0.003562	7.192	12.555
residual_harmonic_gru_sequence_sparse_rcim_global	0.003287	0.003611	7.282	12.461
residual_harmonic_lstm_sequence_dense240_global	0.006247	0.008597	14.061	20.066
residual_harmonic_gru_sequence_dense240_global	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_Fw	0.006983	0.009275	15.722	19.139
residual_harmonic_lstm_sequence_dense240_Fw	0.007042	0.009370	15.868	19.358
residual_harmonic_lstm_sequence_dense360_Fw	0.007731	0.010235	17.430	21.285
residual_harmonic_gru_sequence_dense360_global	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_Fw	0.007869	0.010574	17.740	20.918
residual_harmonic_lstm_sequence_dense360_global	0.008760	0.012992	19.759	24.068

Wave 3.1 Offset-Aware Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
sequential_residual_offset_probe_global	0.003425	0.003839	7.599	12.346

Wave 3.2 Harmonic-Offset Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2f_bis_harmonic_residual_offset_global	0.003255	0.003547	7.224	12.114
track2f_bis_clean_sequential_residual_offset_global	0.003420	0.003841	7.590	13.476

Wave 3.3 Curve-Aware Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_raw_centered_shape_global	0.003248	0.003556	7.218	12.268
track2g_curve_aware_raw_offset_global	0.003359	0.003630	7.429	12.737
track2g_curve_aware_pointwise_control_global	0.003499	0.003784	7.771	13.513
track2g_curve_aware_full_curve_composite_global	0.085677	0.085733	188.514	319.710

Wave 5.2 series Robust-Loss Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_smooth_l1_robust_global	0.003222	0.003507	7.161	12.021
track2h_mae_robust_global	0.003235	0.003544	7.189	12.856
track2h_log_cosh_robust_global	0.003515	0.003825	7.792	14.729

Wave 5.2 series Quantile Probabilistic Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_gaussian_nll_global	0.002951	0.003246	6.524	15.237
track2h_quantile_p10_p50_p90_global	0.003188	0.003469	7.059	12.765

Wave 5.2 series Mixture Density Heads Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_mdn_k2_global	0.003263	0.003582	7.261	14.354
track2h_mdn_k3_global	0.003415	0.003713	7.594	15.012

Wave 4.4 Latent-State Hysteresis Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_l_causal_tcn_offset_residual_global	0.003267	0.003715	7.241	12.649
track2h_l_gru_offset_residual_global	0.087641	0.087704	192.524	320.620

Wave 5.1 Harmonic-Prior Residual Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave3_harmonic_prior_residual_pointwise_control_global	0.003279	0.003550	7.289	14.109
wave3_harmonic_prior_residual_smooth_l1_structured_global	0.087837	0.087893	192.823	321.880

Wave 5.2B Offset And Harmonic Guided Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_global	0.080306	0.080373	176.296	312.921

Polished Model-Development Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_tree_global	0.066315	0.066379	145.420	257.447
polished_harmonic_regression_global	0.081589	0.081641	179.273	315.612
polished_wave4_2_gaussian_nll_global	0.082468	0.082526	181.304	319.127
polished_wave4_3_mixture_density_k2_global	0.082547	0.082604	181.584	320.574
polished_wave4_2_quantile_p10_p50_p90_global	0.083629	0.083685	183.639	320.063
polished_wave3_3_raw_offset_curve_aware_global	0.084156	0.084216	185.372	318.550
polished_wave4_1_mae_robust_loss_global	0.084350	0.084403	185.643	318.504
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	0.084916	0.084974	186.350	319.895
polished_wave3_3_curve_aware_pointwise_control_global	0.085063	0.085120	187.090	319.094
polished_wave5_1_harmonic_prior_pointwise_control_global	0.085374	0.085428	187.413	317.760
polished_wave3_3_full_curve_composite_global	0.085284	0.085341	187.596	317.599
polished_wave3_2_harmonic_residual_offset_global	0.085263	0.085320	187.765	318.467
polished_wave3_3_raw_centered_shape_curve_aware_global	0.085695	0.085750	188.662	318.403
polished_gru_sequence_global	0.086025	0.086083	189.005	317.427
polished_wave4_1_smooth_l1_robust_loss_global	0.086037	0.086090	189.280	317.088
polished_periodic_temporal_convolution_global	0.086168	0.086235	189.292	316.954
polished_temporal_convolution_global	0.086158	0.086217	189.546	315.846
polished_periodic_gru_sequence_global	0.086508	0.086565	190.171	321.113
polished_residual_harmonic_gru_sequence_sparse_rcim_global	0.086770	0.086811	190.773	318.810
polished_wave4_1_log_cosh_robust_loss_global	0.086857	0.086909	190.908	322.502
polished_feedforward_global	0.087265	0.087342	191.515	323.389
polished_lstm_sequence_global	0.087177	0.087231	191.649	320.837
polished_residual_harmonic_lstm_sequence_dense360_global	0.087693	0.088041	192.988	320.637
polished_wave4_3_mixture_density_k3_global	0.087895	0.087945	193.020	325.272
polished_residual_harmonic_gru_sequence_dense360_global	0.087694	0.088076	193.036	320.668
polished_residual_harmonic_mlp_global	0.088016	0.088064	193.130	317.519
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	0.087927	0.087967	193.386	321.315
polished_wave4_4_gru_latent_offset_residual_global	0.088207	0.088272	193.741	319.288
polished_wave4_4_causal_tcn_latent_offset_residual_global	0.088407	0.088466	194.038	321.840
polished_wave3_2_clean_sequential_residual_offset_global	0.088402	0.088457	194.459	318.670
polished_periodic_lstm_sequence_global	0.088836	0.088902	194.827	324.051
polished_residual_harmonic_lstm_sequence_dense240_global	0.088681	0.088799	195.120	322.794
polished_wave3_1_sequential_residual_offset_probe_global	0.088966	0.089029	195.572	320.192
polished_periodic_mlp_global	0.089219	0.089281	195.795	320.557
polished_residual_harmonic_gru_sequence_dense240_global	0.089108	0.089219	195.824	324.871

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_periodic_mlp_harmonic_global	0.089445	0.089501	196.443	321.277

Backward Comparison

Retuned Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Bw	0.002766	0.003300	5.398	12.280
rcim_retuned_ET19_Bw	0.003441	0.004028	7.021	15.287
rcim_retuned_ERT19_Bw	0.003551	0.004161	7.269	13.187
rcim_retuned_RF19_Bw	0.003649	0.004256	7.543	15.083
paper_retuned_best_Bw	0.003675	0.004284	7.572	15.645
rcim_retuned_SVM19_Bw	0.004016	0.004599	8.813	17.215
rcim_retuned_DT19_Bw	0.004578	0.005169	9.728	19.601
rcim_retuned_HGBM19_Bw	0.004683	0.005301	9.978	17.712
rcim_retuned_LGBM19_Bw	0.008105	0.008655	18.057	35.748
rcim_retuned_ELM19_Bw	0.008917	0.009518	20.169	51.896
rcim_retuned_XGBM19_Bw	0.010679	0.011209	24.184	48.082
rcim_retuned_MLP19_Bw	0.019115	0.023025	44.141	88.991

RCIM Model-Bank Reproduction Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
SVM19_Bw	0.004822	0.005116	10.864	25.533
track1_best_Bw	0.005027	0.005212	11.860	48.106
LGBM19_Bw	0.005037	0.005231	11.880	48.106
GBM19_Bw	0.005180	0.005363	12.252	49.984
DT19_Bw	0.005226	0.005409	12.359	48.860
ERT19_Bw	0.005258	0.005442	12.434	51.665
RF19_Bw	0.005392	0.005584	12.731	55.740
ET19_Bw	0.006273	0.006520	14.314	48.624
HGBM19_Bw	0.006619	0.006834	15.494	53.982
XGBM19_Bw	0.007991	0.008195	18.722	59.067
ELM19_Bw	0.010071	0.010486	23.034	52.159
MLP19_Bw	0.036681	0.044921	86.544	225.831

Wave 1 Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_Bw	0.003258	0.003651	7.051	14.116
tree_global	0.003290	0.003702	7.118	13.703
periodic_mlp_Bw	0.003574	0.004006	7.807	13.970
periodic_mlp_global	0.003610	0.004030	7.879	13.661
residual_harmonic_mlp_global	0.003639	0.004044	7.967	14.489
feedforward_Bw	0.003644	0.004067	7.977	14.284
residual_harmonic_mlp_Bw	0.003672	0.004089	8.046	14.973
harmonic_regression_Bw	0.003745	0.004173	8.171	15.684
feedforward_global	0.003741	0.004188	8.186	15.259
harmonic_regression_global	0.018006	0.018276	41.275	83.603

Wave 2.1 Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
periodic_gru_sequence_global	0.002630	0.002876	6.010	12.693
periodic_lstm_sequence_Bw	0.002625	0.002877	6.013	15.382
periodic_lstm_sequence_global	0.002689	0.002956	6.098	14.674
lstm_sequence_global	0.003515	0.003944	7.666	12.502
lstm_sequence_Bw	0.003555	0.003985	7.767	14.507
gru_sequence_Bw	0.003626	0.004082	7.907	12.900
periodic_temporal_convolution_global	0.003604	0.003948	7.935	13.804
gru_sequence_global	0.003637	0.004080	7.946	14.580
periodic_temporal_convolution_Bw	0.003628	0.003987	7.979	13.839
temporal_convolution_Bw	0.003742	0.004166	8.184	13.908
temporal_convolution_global	0.003994	0.004438	8.798	18.339

Wave 2.3 Residual Harmonic Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_lstm_sequence_sparse_rcim_Bw	0.003440	0.003793	7.510	13.160
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003491	0.003876	7.627	13.540
residual_harmonic_gru_sequence_sparse_rcim_Bw	0.003502	0.003857	7.654	12.829
residual_harmonic_gru_sequence_sparse_rcim_global	0.003584	0.003957	7.859	14.946
residual_harmonic_lstm_sequence_dense240_global	0.006591	0.008932	14.858	21.728
residual_harmonic_gru_sequence_dense240_global	0.006786	0.009221	15.282	20.990
residual_harmonic_lstm_sequence_dense240_Bw	0.007367	0.009945	16.660	22.200
residual_harmonic_gru_sequence_dense360_global	0.008181	0.011621	18.502	23.968
residual_harmonic_lstm_sequence_dense360_global	0.008860	0.013061	20.073	23.940
residual_harmonic_gru_sequence_dense240_Bw	0.008984	0.012987	20.358	26.370
residual_harmonic_gru_sequence_dense360_Bw	0.009370	0.013165	21.267	25.901
residual_harmonic_lstm_sequence_dense360_Bw	0.010268	0.014769	23.355	29.779

Wave 3.1 Offset-Aware Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
sequential_residual_offset_probe_global	0.003646	0.004080	7.981	14.394

Wave 3.2 Harmonic-Offset Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2f_bis_clean_sequential_residual_offset_global	0.003624	0.004058	7.918	12.339
track2f_bis_harmonic_residual_offset_global	0.003805	0.004120	8.354	14.523

Wave 3.3 Curve-Aware Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_raw_centered_shape_global	0.003449	0.003808	7.572	14.065
track2g_curve_aware_raw_offset_global	0.003559	0.003881	7.831	13.812
track2g_curve_aware_pointwise_control_global	0.003658	0.004016	8.051	15.707
track2g_curve_aware_full_curve_composite_global	0.005829	0.006210	12.816	27.371

Wave 5.2 series Robust-Loss Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_log_cosh_robust_global	0.003481	0.003814	7.603	13.596
track2h_mae_robust_global	0.003568	0.003886	7.818	14.652
track2h_smooth_l1_robust_global	0.003612	0.003931	7.918	14.825

Wave 5.2 series Quantile Probabilistic Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_gaussian_nll_global	0.003068	0.003372	6.627	15.928
track2h_quantile_p10_p50_p90_global	0.003563	0.003909	7.816	15.378

Wave 5.2 series Mixture Density Heads Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_mdn_k3_global	0.003701	0.004023	8.129	16.261
track2h_mdn_k2_global	0.003735	0.004073	8.194	15.436

Wave 4.4 Latent-State Hysteresis Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_l_causal_tcn_offset_residual_global	0.003478	0.003940	7.555	13.771
track2h_l_gru_offset_residual_global	0.008530	0.008976	18.741	40.030

Wave 5.1 Harmonic-Prior Residual Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave3_harmonic_prior_residual_pointwise_control_global	0.003604	0.003959	7.906	14.071
wave3_harmonic_prior_residual_smooth_l1_structured_global	0.008517	0.008877	18.660	41.964

Wave 5.2B Offset And Harmonic Guided Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_global	0.004781	0.005129	10.705	26.696

Polished Model-Development Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_harmonic_regression_global	0.004331	0.004726	9.481	21.909
polished_wave4_1_mae_robust_loss_global	0.004734	0.005066	10.371	22.678
polished_wave4_2_gaussian_nll_global	0.004723	0.005058	10.446	22.222
polished_wave4_3_mixture_density_k2_global	0.004832	0.005146	10.763	23.393
polished_wave3_3_raw_offset_curve_aware_global	0.004985	0.005314	11.054	23.547
polished_wave3_2_harmonic_residual_offset_global	0.005272	0.005623	11.611	25.054
polished_wave4_2_quantile_p10_p50_p90_global	0.005455	0.005807	11.933	25.004
polished_wave3_3_full_curve_composite_global	0.005763	0.006116	12.644	28.205
polished_wave3_3_curve_aware_pointwise_control_global	0.005754	0.006094	12.647	25.192
polished_wave4_1_smooth_l1_robust_loss_global	0.005958	0.006297	13.003	32.185
polished_wave3_3_raw_centered_shape_curve_aware_global	0.005942	0.006277	13.131	29.183
polished_temporal_convolution_global	0.006542	0.007000	14.321	32.961
polished_wave5_1_harmonic_prior_pointwise_control_global	0.006599	0.006949	14.435	33.050
polished_periodic_temporal_convolution_global	0.006842	0.007285	14.811	35.825
polished_residual_harmonic_gru_sequence_sparse_rcim_global	0.006776	0.007157	14.838	30.081
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	0.006762	0.007113	14.904	33.497
polished_gru_sequence_global	0.006814	0.007285	14.950	30.038
polished_wave4_1_log_cosh_robust_loss_global	0.007318	0.007648	16.063	36.570
polished_periodic_gru_sequence_global	0.007336	0.007715	16.264	32.319
polished_lstm_sequence_global	0.007449	0.007853	16.308	41.180
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	0.007575	0.007936	16.542	44.368
polished_feedforward_global	0.008231	0.008800	17.869	42.849
polished_wave3_2_clean_sequential_residual_offset_global	0.008207	0.008634	18.039	41.136
polished_residual_harmonic_lstm_sequence_dense360_global	0.008233	0.009936	18.219	37.015
polished_wave4_3_mixture_density_k3_global	0.008362	0.008683	18.302	42.032
polished_wave4_4_gru_latent_offset_residual_global	0.008451	0.008942	18.421	39.768
polished_residual_harmonic_gru_sequence_dense360_global	0.008335	0.010156	18.494	35.068
polished_residual_harmonic_lstm_sequence_dense240_global	0.008410	0.009192	18.525	43.908
polished_residual_harmonic_mlp_global	0.008685	0.009067	18.784	45.907
polished_wave4_4_causal_tcn_latent_offset_residual_global	0.008738	0.009177	18.907	46.002
polished_wave3_1_sequential_residual_offset_probe_global	0.008771	0.009248	19.185	44.010
polished_periodic_mlp_harmonic_global	0.009199	0.009571	19.896	48.481
polished_residual_harmonic_gru_sequence_dense240_global	0.009240	0.009932	20.241	44.364
polished_periodic_lstm_sequence_global	0.009636	0.010017	20.812	46.697
polished_periodic_mlp_global	0.009681	0.010115	20.915	51.053

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_tree_global	0.015215	0.015523	34.609	61.506

Global Model Direction Breakdown

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
feedforward_global	forward	0.003550	0.003979	7.916	16.195
feedforward_global	backward	0.003741	0.004188	8.186	15.259
feedforward_global	combined	0.003646	0.004083	8.051	15.583
gru_sequence_global	forward	0.003546	0.003975	7.869	14.344
gru_sequence_global	backward	0.003637	0.004080	7.946	14.580
gru_sequence_global	combined	0.003591	0.004028	7.907	14.389
harmonic_regression_global	forward	0.018314	0.018547	41.749	79.404
harmonic_regression_global	backward	0.018006	0.018276	41.275	83.603
harmonic_regression_global	combined	0.018160	0.018412	41.512	82.048
lstm_sequence_global	forward	0.003445	0.003863	7.642	12.032
lstm_sequence_global	backward	0.003515	0.003944	7.666	12.502
lstm_sequence_global	combined	0.003480	0.003903	7.654	12.430
periodic_gru_sequence_global	forward	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_global	backward	0.002630	0.002876	6.010	12.693
periodic_gru_sequence_global	combined	0.002704	0.002949	6.139	13.200
periodic_lstm_sequence_global	forward	0.002726	0.002959	6.142	14.092
periodic_lstm_sequence_global	backward	0.002689	0.002956	6.098	14.674
periodic_lstm_sequence_global	combined	0.002707	0.002958	6.120	14.717
periodic_mlp_global	forward	0.003380	0.003791	7.493	14.349
periodic_mlp_global	backward	0.003610	0.004030	7.879	13.661
periodic_mlp_global	combined	0.003495	0.003910	7.686	14.284
periodic_temporal_convolution_global	forward	0.003407	0.003724	7.581	14.480
periodic_temporal_convolution_global	backward	0.003604	0.003948	7.935	13.804
periodic_temporal_convolution_global	combined	0.003506	0.003836	7.758	14.308
polished_feedforward_global	forward	0.087265	0.087342	191.515	323.389
polished_feedforward_global	backward	0.008231	0.008800	17.869	42.849
polished_feedforward_global	combined	0.047748	0.048071	104.692	316.205
polished_gru_sequence_global	forward	0.086025	0.086083	189.005	317.427
polished_gru_sequence_global	backward	0.006814	0.007285	14.950	30.038
polished_gru_sequence_global	combined	0.046419	0.046684	101.977	306.679
polished_harmonic_regression_global	forward	0.081589	0.081641	179.273	315.612
polished_harmonic_regression_global	backward	0.004331	0.004726	9.481	21.909
polished_harmonic_regression_global	combined	0.042960	0.043184	94.377	300.154
polished_lstm_sequence_global	forward	0.087177	0.087231	191.649	320.837
polished_lstm_sequence_global	backward	0.007449	0.007853	16.308	41.180
polished_lstm_sequence_global	combined	0.047313	0.047542	103.978	312.066
polished_periodic_gru_sequence_global	forward	0.086508	0.086565	190.171	321.113
polished_periodic_gru_sequence_global	backward	0.007336	0.007715	16.264	32.319
polished_periodic_gru_sequence_global	combined	0.046922	0.047140	103.217	308.291

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_periodic_lstm_sequence_global	forward	0.088836	0.088902	194.827	324.051
polished_periodic_lstm_sequence_global	backward	0.009636	0.010017	20.812	46.697
polished_periodic_lstm_sequence_global	combined	0.049236	0.049460	107.819	316.139
polished_periodic_mlp_global	forward	0.089219	0.089281	195.795	320.557
polished_periodic_mlp_global	backward	0.009681	0.010115	20.915	51.053
polished_periodic_mlp_global	combined	0.049450	0.049698	108.355	315.126
polished_periodic_mlp_harmonic_global	forward	0.089445	0.089501	196.443	321.277
polished_periodic_mlp_harmonic_global	backward	0.009199	0.009571	19.896	48.481
polished_periodic_mlp_harmonic_global	combined	0.049322	0.049536	108.169	315.547
polished_periodic_temporal_convolution_global	forward	0.086168	0.086235	189.292	316.954
polished_periodic_temporal_convolution_global	backward	0.006842	0.007285	14.811	35.825
polished_periodic_temporal_convolution_global	combined	0.046505	0.046760	102.052	307.929
polished_residual_harmonic_gru_sequence_dense240_global	forward	0.089108	0.089219	195.824	324.871
polished_residual_harmonic_gru_sequence_dense240_global	backward	0.009240	0.009932	20.241	44.364
polished_residual_harmonic_gru_sequence_dense240_global	combined	0.049174	0.049576	108.033	316.565
polished_residual_harmonic_gru_sequence_dense360_global	forward	0.087694	0.088076	193.036	320.668
polished_residual_harmonic_gru_sequence_dense360_global	backward	0.008335	0.010156	18.494	35.068
polished_residual_harmonic_gru_sequence_dense360_global	combined	0.048014	0.049116	105.765	307.120
polished_residual_harmonic_gru_sequence_sparse_rcim_global	forward	0.086770	0.086811	190.773	318.810
polished_residual_harmonic_gru_sequence_sparse_rcim_global	backward	0.006776	0.007157	14.838	30.081
polished_residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.046773	0.046984	102.806	307.471
polished_residual_harmonic_lstm_sequence_dense240_global	forward	0.088681	0.088799	195.120	322.794
polished_residual_harmonic_lstm_sequence_dense240_global	backward	0.008410	0.009192	18.525	43.908
polished_residual_harmonic_lstm_sequence_dense240_global	combined	0.048546	0.048996	106.822	314.965
polished_residual_harmonic_lstm_sequence_dense360_global	forward	0.087693	0.088041	192.988	320.637
polished_residual_harmonic_lstm_sequence_dense360_global	backward	0.008233	0.009936	18.219	37.015
polished_residual_harmonic_lstm_sequence_dense360_global	combined	0.047963	0.048989	105.603	309.044
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	forward	0.087927	0.087967	193.386	321.315
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	backward	0.007575	0.007936	16.542	44.368
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.047751	0.047952	104.964	314.106
polished_residual_harmonic_mlp_global	forward	0.088016	0.088064	193.130	317.519
polished_residual_harmonic_mlp_global	backward	0.008685	0.009067	18.784	45.907
polished_residual_harmonic_mlp_global	combined	0.048350	0.048566	105.957	308.744
polished_temporal_convolution_global	forward	0.086158	0.086217	189.546	315.846

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_temporal_convolution_global	backward	0.006542	0.007000	14.321	32.961
polished_temporal_convolution_global	combined	0.046350	0.046608	101.934	302.803
polished_tree_global	forward	0.066315	0.066379	145.420	257.447
polished_tree_global	backward	0.015215	0.015523	34.609	61.506
polished_tree_global	combined	0.040765	0.040951	90.014	243.826
polished_wave3_1_sequential_residual_offset_probe_global	forward	0.088966	0.089029	195.572	320.192
polished_wave3_1_sequential_residual_offset_probe_global	backward	0.008771	0.009248	19.185	44.010
polished_wave3_1_sequential_residual_offset_probe_global	combined	0.048869	0.049139	107.379	313.017
polished_wave3_2_clean_sequential_residual_offset_global	forward	0.088402	0.088457	194.459	318.670
polished_wave3_2_clean_sequential_residual_offset_global	backward	0.008207	0.008634	18.039	41.136
polished_wave3_2_clean_sequential_residual_offset_global	combined	0.048305	0.048545	106.249	310.064
polished_wave3_2_harmonic_residual_offset_global	forward	0.085263	0.085320	187.765	318.467
polished_wave3_2_harmonic_residual_offset_global	backward	0.005272	0.005623	11.611	25.054
polished_wave3_2_harmonic_residual_offset_global	combined	0.045267	0.045471	99.688	303.092
polished_wave3_3_curve_aware_pointwise_control_global	forward	0.085063	0.085120	187.090	319.094
polished_wave3_3_curve_aware_pointwise_control_global	backward	0.005754	0.006094	12.647	25.192
polished_wave3_3_curve_aware_pointwise_control_global	combined	0.045409	0.045607	99.868	303.607
polished_wave3_3_full_curve_composite_global	forward	0.085284	0.085341	187.596	317.599
polished_wave3_3_full_curve_composite_global	backward	0.005763	0.006116	12.644	28.205
polished_wave3_3_full_curve_composite_global	combined	0.045523	0.045729	100.120	303.120
polished_wave3_3_raw_centered_shape_curve_aware_global	forward	0.085695	0.085750	188.662	318.403
polished_wave3_3_raw_centered_shape_curve_aware_global	backward	0.005942	0.006277	13.131	29.183
polished_wave3_3_raw_centered_shape_curve_aware_global	combined	0.045819	0.046013	100.896	303.810
polished_wave3_3_raw_offset_curve_aware_global	forward	0.084156	0.084216	185.372	318.550
polished_wave3_3_raw_offset_curve_aware_global	backward	0.004985	0.005314	11.054	23.547
polished_wave3_3_raw_offset_curve_aware_global	combined	0.044570	0.044765	98.213	297.738
polished_wave4_1_log_cosh_robust_loss_global	forward	0.086857	0.086909	190.908	322.502
polished_wave4_1_log_cosh_robust_loss_global	backward	0.007318	0.007648	16.063	36.570
polished_wave4_1_log_cosh_robust_loss_global	combined	0.047088	0.047279	103.486	309.260
polished_wave4_1_mae_robust_loss_global	forward	0.084350	0.084403	185.643	318.504
polished_wave4_1_mae_robust_loss_global	backward	0.004734	0.005066	10.371	22.678
polished_wave4_1_mae_robust_loss_global	combined	0.044542	0.044734	98.007	302.800
polished_wave4_1_smooth_l1_robust_loss_global	forward	0.086037	0.086090	189.280	317.088
polished_wave4_1_smooth_l1_robust_loss_global	backward	0.005958	0.006297	13.003	32.185

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_wave4_1_smooth_l1_robust_loss_global	combined	0.045997	0.046194	101.141	303.876
polished_wave4_2_gaussian_nll_global	forward	0.082468	0.082526	181.304	319.127
polished_wave4_2_gaussian_nll_global	backward	0.004723	0.005058	10.446	22.222
polished_wave4_2_gaussian_nll_global	combined	0.043595	0.043792	95.875	301.197
polished_wave4_2_quantile_p10_p50_p90_global	forward	0.083629	0.083685	183.639	320.063
polished_wave4_2_quantile_p10_p50_p90_global	backward	0.005455	0.005807	11.933	25.004
polished_wave4_2_quantile_p10_p50_p90_global	combined	0.044542	0.044746	97.786	303.539
polished_wave4_3_mixture_density_k2_global	forward	0.082547	0.082604	181.584	320.574
polished_wave4_3_mixture_density_k2_global	backward	0.004832	0.005146	10.763	23.393
polished_wave4_3_mixture_density_k2_global	combined	0.043690	0.043875	96.173	303.061
polished_wave4_3_mixture_density_k3_global	forward	0.087895	0.087945	193.020	325.272
polished_wave4_3_mixture_density_k3_global	backward	0.008362	0.008683	18.302	42.032
polished_wave4_3_mixture_density_k3_global	combined	0.048129	0.048314	105.661	313.475
polished_wave4_4_causal_tcn_latent_offset_residual_global	forward	0.088407	0.088466	194.038	321.840
polished_wave4_4_causal_tcn_latent_offset_residual_global	backward	0.008738	0.009177	18.907	46.002
polished_wave4_4_causal_tcn_latent_offset_residual_global	combined	0.048573	0.048822	106.472	315.489
polished_wave4_4_gru_latent_offset_residual_global	forward	0.088207	0.088272	193.741	319.288
polished_wave4_4_gru_latent_offset_residual_global	backward	0.008451	0.008942	18.421	39.768
polished_wave4_4_gru_latent_offset_residual_global	combined	0.048329	0.048607	106.081	309.047
polished_wave5_1_harmonic_prior_pointwise_control_global	forward	0.085374	0.085428	187.413	317.760
polished_wave5_1_harmonic_prior_pointwise_control_global	backward	0.006599	0.006949	14.435	33.050
polished_wave5_1_harmonic_prior_pointwise_control_global	combined	0.045987	0.046189	100.924	304.995
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	forward	0.084916	0.084974	186.350	319.895
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	backward	0.006762	0.007113	14.904	33.497
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	combined	0.045839	0.046043	100.627	307.281
residual_harmonic_gru_sequence_dense240_global	forward	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_global	backward	0.006786	0.009221	15.282	20.990
residual_harmonic_gru_sequence_dense240_global	combined	0.006660	0.009090	15.007	20.940
residual_harmonic_gru_sequence_dense360_global	forward	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_global	backward	0.008181	0.011621	18.502	23.968
residual_harmonic_gru_sequence_dense360_global	combined	0.008012	0.011416	18.090	23.559
residual_harmonic_gru_sequence_sparse_rcim_global	forward	0.003287	0.003611	7.282	12.461
residual_harmonic_gru_sequence_sparse_rcim_global	backward	0.003584	0.003957	7.859	14.946

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.003435	0.003784	7.571	13.869
residual_harmonic_lstm_sequence_dense240_global	forward	0.006247	0.008597	14.061	20.066
residual_harmonic_lstm_sequence_dense240_global	backward	0.006591	0.008932	14.858	21.728
residual_harmonic_lstm_sequence_dense240_global	combined	0.006419	0.008765	14.460	20.628
residual_harmonic_lstm_sequence_dense360_global	forward	0.008760	0.012992	19.759	24.068
residual_harmonic_lstm_sequence_dense360_global	backward	0.008860	0.013061	20.073	23.940
residual_harmonic_lstm_sequence_dense360_global	combined	0.008810	0.013026	19.916	24.052
residual_harmonic_lstm_sequence_sparse_rcim_global	forward	0.003245	0.003562	7.192	12.555
residual_harmonic_lstm_sequence_sparse_rcim_global	backward	0.003491	0.003876	7.627	13.540
residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.003368	0.003719	7.409	13.041
residual_harmonic_mlp_global	forward	0.003649	0.004051	8.123	14.713
residual_harmonic_mlp_global	backward	0.003639	0.004044	7.967	14.489
residual_harmonic_mlp_global	combined	0.003644	0.004048	8.045	14.717
sequential_residual_offset_probe_global	forward	0.003425	0.003839	7.599	12.346
sequential_residual_offset_probe_global	backward	0.003646	0.004080	7.981	14.394
sequential_residual_offset_probe_global	combined	0.003536	0.003959	7.790	13.355
temporal_convolution_global	forward	0.003508	0.003928	7.792	14.045
temporal_convolution_global	backward	0.003994	0.004438	8.798	18.339
temporal_convolution_global	combined	0.003751	0.004183	8.295	17.247
track2f_bis_clean_sequential_residual_offset_global	forward	0.003420	0.003841	7.590	13.476
track2f_bis_clean_sequential_residual_offset_global	backward	0.003624	0.004058	7.918	12.339
track2f_bis_clean_sequential_residual_offset_global	combined	0.003522	0.003950	7.754	13.166
track2f_bis_harmonic_residual_offset_global	forward	0.003255	0.003547	7.224	12.114
track2f_bis_harmonic_residual_offset_global	backward	0.003805	0.004120	8.354	14.523
track2f_bis_harmonic_residual_offset_global	combined	0.003530	0.003833	7.789	13.523
track2g_curve_aware_full_curve_composite_global	forward	0.085677	0.085733	188.514	319.710
track2g_curve_aware_full_curve_composite_global	backward	0.005829	0.006210	12.816	27.371
track2g_curve_aware_full_curve_composite_global	combined	0.045753	0.045971	100.665	304.044
track2g_curve_aware_pointwise_control_global	forward	0.003499	0.003784	7.771	13.513
track2g_curve_aware_pointwise_control_global	backward	0.003658	0.004016	8.051	15.707
track2g_curve_aware_pointwise_control_global	combined	0.003578	0.003900	7.911	15.536
track2g_curve_aware_raw_centered_shape_global	forward	0.003248	0.003556	7.218	12.268
track2g_curve_aware_raw_centered_shape_global	backward	0.003449	0.003808	7.572	14.065
track2g_curve_aware_raw_centered_shape_global	combined	0.003348	0.003682	7.395	13.476
track2g_curve_aware_raw_offset_global	forward	0.003359	0.003630	7.429	12.737

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_raw_offset_global	backward	0.003559	0.003881	7.831	13.812
track2g_curve_aware_raw_offset_global	combined	0.003459	0.003755	7.630	13.330
track2h_gaussian_nll_global	forward	0.002951	0.003246	6.524	15.237
track2h_gaussian_nll_global	backward	0.003068	0.003372	6.627	15.928
track2h_gaussian_nll_global	combined	0.003009	0.003309	6.576	15.849
track2h_l_causal_tcn_offset_residual_global	forward	0.003267	0.003715	7.241	12.649
track2h_l_causal_tcn_offset_residual_global	backward	0.003478	0.003940	7.555	13.771
track2h_l_causal_tcn_offset_residual_global	combined	0.003372	0.003827	7.398	13.454
track2h_l_gru_offset_residual_global	forward	0.087641	0.087704	192.524	320.620
track2h_l_gru_offset_residual_global	backward	0.008530	0.008976	18.741	40.030
track2h_l_gru_offset_residual_global	combined	0.048086	0.048340	105.632	309.262
track2h_log_cosh_robust_global	forward	0.003515	0.003825	7.792	14.729
track2h_log_cosh_robust_global	backward	0.003481	0.003814	7.603	13.596
track2h_log_cosh_robust_global	combined	0.003498	0.003819	7.697	14.405
track2h_mae_robust_global	forward	0.003235	0.003544	7.189	12.856
track2h_mae_robust_global	backward	0.003568	0.003886	7.818	14.652
track2h_mae_robust_global	combined	0.003401	0.003715	7.504	13.873
track2h_mdn_k2_global	forward	0.003263	0.003582	7.261	14.354
track2h_mdn_k2_global	backward	0.003735	0.004073	8.194	15.436
track2h_mdn_k2_global	combined	0.003499	0.003828	7.727	14.956
track2h_mdn_k3_global	forward	0.003415	0.003713	7.594	15.012
track2h_mdn_k3_global	backward	0.003701	0.004023	8.129	16.261
track2h_mdn_k3_global	combined	0.003558	0.003868	7.861	15.990
track2h_quantile_p10_p50_p90_global	forward	0.003188	0.003469	7.059	12.765
track2h_quantile_p10_p50_p90_global	backward	0.003563	0.003909	7.816	15.378
track2h_quantile_p10_p50_p90_global	combined	0.003375	0.003689	7.438	14.851
track2h_smooth_l1_robust_global	forward	0.003222	0.003507	7.161	12.021
track2h_smooth_l1_robust_global	backward	0.003612	0.003931	7.918	14.825
track2h_smooth_l1_robust_global	combined	0.003417	0.003719	7.539	14.332
tree_global	forward	0.002998	0.003364	6.590	11.601
tree_global	backward	0.003290	0.003702	7.118	13.703
tree_global	combined	0.003144	0.003533	6.854	13.314
wave3_harmonic_prior_residual_pointwise_control_global	forward	0.003279	0.003550	7.289	14.109
wave3_harmonic_prior_residual_pointwise_control_global	backward	0.003604	0.003959	7.906	14.071
wave3_harmonic_prior_residual_pointwise_control_global	combined	0.003442	0.003755	7.597	14.110
wave3_harmonic_prior_residual_smooth_l1_structured_global	forward	0.087837	0.087893	192.823	321.880
wave3_harmonic_prior_residual_smooth_l1_structured_global	backward	0.008517	0.008877	18.660	41.964
wave3_harmonic_prior_residual_smooth_l1_structured_global	combined	0.048177	0.048385	105.742	313.347
wave52b_offset_centered_shape_harmonic_global	forward	0.080306	0.080373	176.296	312.921
wave52b_offset_centered_shape_harmonic_global	backward	0.004781	0.005129	10.705	26.696
wave52b_offset_centered_shape_harmonic_global	combined	0.042544	0.042751	93.500	295.192

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-05-18-55-59__track2_full_directional_family_matrix_track2_dataset_surface_simplified_dataset_global_2026_07_04\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-05-18-55-59__track2_full_directional_family_matrix_track2_dataset_surface_simplified_dataset_global_2026_07_04\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional paper-reference, Wave 1, and Wave 2.1 models are never evaluated on the opposite direction. Global Wave models remain valid on both directions and are therefore shown in the directional sections and again in the global breakdown. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.